Tooth Morphology + Anticipatory Guidance
OMFS 721 – Pediatric Dentistry

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Mission: To advance the science and art of health through education, service, scholarship and social accountability

Quiz Time!

What are we covering today?

- Primary tooth morphology basics (plus some tips and tricks)
- Anticipatory guidance fundamentals
- We will eventually be incorporating everything together to really understand how anticipatory guidance works

General Trends in Teeth

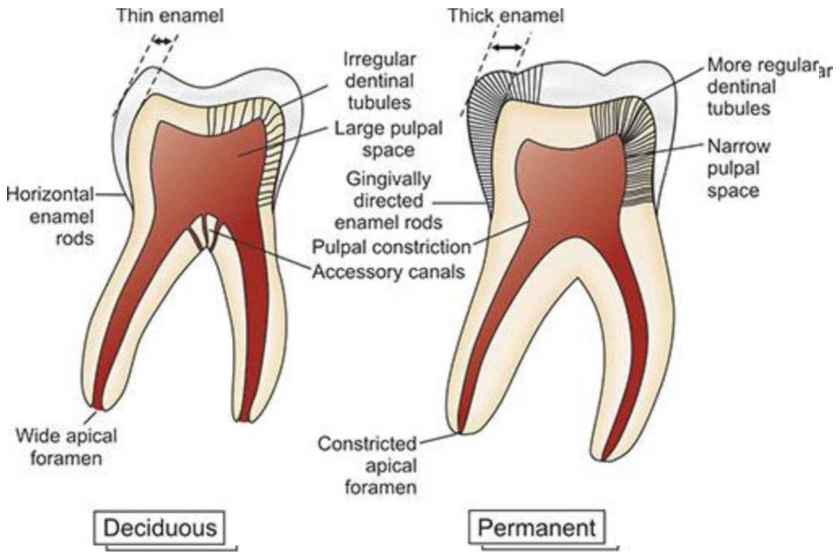
- Lower teeth usually comes in first
  - Exfoliation typically **follows the order** primary teeth came in
  - Also true of speed: if teeth come *in* earlier then they fall *out* earlier
- One side may develop faster than the other
- **Rule of 6's**
  - Every **six years** a new permanent molar erupts
  - Useful goalposts for tooth eruption dates

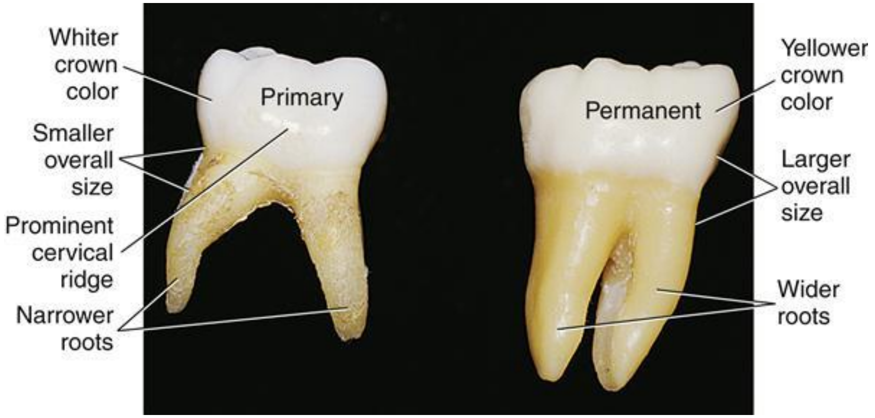
Primary Tooth Morphology

- Primary teeth are smaller and rounder
  - Contacts are wider/broader
- Less enamel
  - ...but also less dentin
  - Teeth are more translucent and whiter

Primary Molars

- Large pulp chamber and horns
  - More likely to pulp out
- Pulp chamber usually has multiple accessory canals
  - Pulpotomy effective!
- Roots are usually flared outwards
  - Room for permanent teeth to develop and erupt
  - Makes extractions challenging





Recognizing primary vs. permanent teeth

Primary Tooth Eruption

Primary Dentition						
	Calcification begins at	Formation complete at	Eruption		Exfoliation	
			Maxillary	Mandibular	Maxillary	Mandibular
Central incisors	4th fetal mo	18-24 mo	6-10 mo	5-8 mo	7-8 y	6-7 y
Lateral incisors	4th fetal mo	18-24 mo	8-12 mo	7-10 mo	8-9 y	7-8 y
Canines	4th fetal mo	30-39 mo	16-20 mo	16-20 mo	11-12 y	9-11 y
First molars	4th fetal mo	24-30 mo	11-18 mo	11-18 mo	9-11 y	10-12 y
Second molars	4th fetal mo	36 mo	20-30 mo	20-30 mo	9-12 y	11-13 y

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Second molars	4th fetal mo	36 mo	20-30 mo	20-30 mo	9-12 y	11-13 y

Usually, having all primary anteriors at ~12mo is a reasonable assumption to make
Recall: what is the AAPD criteria for establishing a dental home?

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"Last to come in, last to go"

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Permanent Tooth Eruption

- Roots typically complete ~3 years after eruption of the tooth
- Crown completes in half the time of eruption date

Permanent Dentition					
	Calcification begins at	Crown (enamel) complete at	Roots complete at	Eruption* Maxillary	Mandibular
Central incisiors	3-4 mo	4-5 y	9-10 y	7-8 y (3)	6-7 y (2)
Lateral incisors	Maxilla: 10-12 mo Mandible: 3-4 mo	4-5 y 4-5 y	11 y 10 y	8-9 y (5)	7-8 y (4)
Canines	4-5 mo	6-7 y	12-15 y	11-12 y (11)	9-11 y (6)
First premolars	18-24 mo	5-6 y	12-13 y	10-11 y (7)	10-12 y (8)
Second premolars	24-30 mo	6-7 y	12-14 y	10-12 y (9)	11-13 y (10)
First molars	Birth	30-36 mo	9-10 y	5.5-7 y (1)	5.5-7 y (1a)
Second molars	30-36 mo	7-8 y	14-16 y	12-14 y (12)	12-14 y (12a)
Third molars	Maxilla: 7-9 y Mandible: 8-10 y			17-30 y (13)	17-30 y (13a)

Permanent Tooth Eruption

Typically, the highlighted teeth begin forming at birth

More susceptible to enamel hypoplasia

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Permanent Tooth Eruption (Rule of 6's)

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Anticipatory Guidance

Anticipatory Guidance

- Definition: the process of providing **practical and developmentally-appropriate** information about children's health to prepare parents for significant physical, emotional, and psychological milestones.
- Kids aged 6-12 yrs old are 4x likely to visit a dentist than a pediatrician
- Topics include:
  - Oral/Dental development and growth
  - Oral Hygiene Instruction (OHI)
  - Diet and Nutrition
  - Habits (non-nutritive sucking)
  - Injury Prevention

Oral Development and Growth

- Regardless of pediatric or adult dentistry, a *good* dental home includes discussion of expectations
- Parents will expect/have questions about upcoming changes
- Baby teeth erupt when they’re ready to erupt
  - Some kiddos will get their teeth faster, some will get them slower
  - The sequences are parallel: first to come in, first to go, etc.
- Parents like asking about ortho
- Timeline of various restorative procedures
  - Ex: How long do I need a space maintainer?

Health Considerations

- Prior to any assessment, patient health should always be evaluated
  - Consults are always key!
- Identifying the patient ASA classification is essential for advanced behavior management
- Airway in particular important in peds because adult emergencies usually result in cardiovascular incidents, children are typically **pulmonary**

OHI Distinctions

- Soft-bristled brushes only!
- Electric toothbrushes not recommended for <3 yrs of age
- Children **SHOULD** use a fluoridated toothpaste:
  - <3 yrs of age: 0.1 mg (**rice-grain or smear** amount)
  - 3-7 yrs of age: 0.25 mg (**pea-sized** amount)

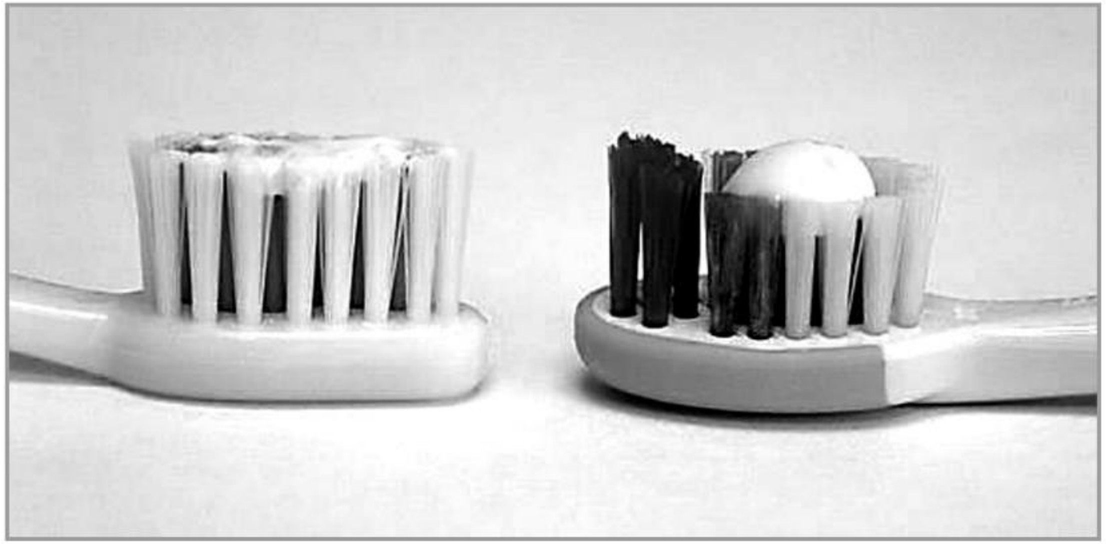


Figure. Comparison of a smear (left) with a pea-sized (right) amount of toothpaste.

OHI Distinction

- Brushing: time spent brushing is better than technique
  - 2 minutes is the recommended time
- Parents should assist with brushing and flossing until the age of 7 **or** when kids are able to tie their own shoelaces
- Type of toothpaste won't matter so long as its fluoridated

Night Time Feeding

- Recall from the Caries Risk Assessment that **bedtime use of the bottle results in high caries risk**.
- Salivary flow is diminished at nighttime and protective factors are lost.
- Infants (although some adults too) are constantly coating their teeth with lactose (present in breast milk, cow's milk, and even formula) which is a carbohydrate.
- Parents should be encouraged to **either brush the teeth after feeding or if not possible, to at least wipe the teeth with a wet cloth**.
  - Gradual dilution of bottle's contents also viable option

Diet and Nutrition

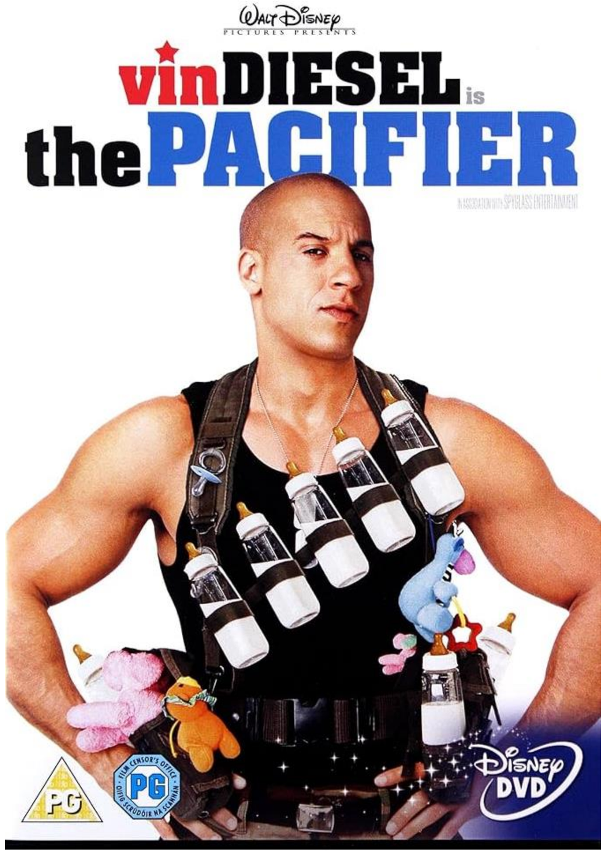
- The Caries Risk Assessment Form indicates children who snack **>3 times a day between meals** are at a **high risk**.
- Children in particular enjoy snacks and many enjoy treats (grandparents, daycares, schools are prime culprits)
- Frequency of carbohydrate exposure and meals increase caries risk.
- Advise parents to reduce frequency of snacking and try to have sweet treats as part of a meal to minimize dental damage.

Sugar

MerseySide

Habits/Non-nutritive Sucking

- Non-nutritive Sucking
  - Thumbs, pacifier, blanket
- Nail Biting
- These are usually **soothing** behaviors and can persist
- Causes orthodontic issues
- Most kids (~60-65%) will quit on their own by the age of 3yrs



Injury Prevention

- Big part of growing up: scrapes and bruises
- Will discuss more in trauma lectures
- General info: kids (typically boys) are highest risk for dental injuries at ages 2-3 yrs
  - People with history of dental trauma are at risk for more trauma
  - 6-7 yr old kids at risk if they play contact sports! Recommend a mouthguard!

Walking Through Anticipatory Guidance

- Effective anticipatory guidance is age-appropriate and concise
- For learning purposes, I divided the content into three life stages: infancy, childhood, and adolescence.
- Remember: Frankl Score is assigned based on context!
  - All infants are by default a Frankl 2, specifically labeled as "Pre-Cooperative"

Infancy (0-3 yrs)

- Establish the dental home
- Is this a first-time parent? Have they or their child ever been to a pediatric dentist?
- Prophylaxis (with a toothbrush)
  - What about brushing at home?
- Radiographs only for emergencies
- Primary tooth eruption sequence/timing
  - Symptoms of teething: Soreness, irritability, drooling, chewing on things

Knee-to-Knee Exam

- Most infants cannot tolerate traditional exam settings
- Knee-to-Knee exam places patient's legs towards parents who will help hold the hands
- Useful time to also demonstrate a strategy for brushing at home if child does not like brushing

Childhood (3-12 yrs)

- Mixed dentition
  - Tooth eruption
  - Early ortho considerations
- Oral habits cont'd
- Sports
  - Age of 7 – recommend mouth guard (permanent anteriors at risk)
- Brushing Guidelines
  - Recommendation is for kids to get assistance with brushing until the age of 7 or if they can tie their own shoelaces

Adolescence

- Late-mixed dentition – talk about changes, ortho, wisdom teeth, etc.
- Sports
- Puberty and Hormonal Changes
  - Tobacco Cessation
  - Drugs and alcohol
  - Sexual/mucosally transmitted diseases
- Teenage Apathy + Mental health issues
  - Medical history will likely change!

Vaping

- Fairly new development
- Nicotine affects adolescent brain development (CDC)
- Adverse lung and cardiovascular effects
- Long-term usage unknown

Questions?